

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I Bhuvaneshwari j(192572141)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Question 1: Compare Type-1 and Type-2 Hypervisors

Objective

Create a virtual machine in VMware Workstation and observe its performance.

Step 1: Open VMware Workstation

1. Launch VMware Workstation.
2. Click Create a New Virtual Machine.

Step 2: Select Configuration

Choose

Typical (Recommended)

Click Next.

Step 3: Select ISO

Choose

Installer disc image file (ISO)

Browse to your Ubuntu or Windows ISO.

Click Next.

Step 4: Enter OS Details

Example

Username: student

Password: student123

Click Next.

Step 5: Name VM

Example

Cloud Server

Location

D:\VMs\CloudServer

Click Next.

Step 6: Configure Disk

60 GB

Store as single file

Click Next.

Step 7: Customize Hardware

Click

Customize Hardware

Set

Memory

8192 MB

Processors

4

Hard Disk

60 GB

Network Adapter

Bridged

CD/DVD

ISO selected

Enable

Processors

✓ Virtualize Intel VT-x/EPT

Click Finish.

Step 8: Start VM

Click

Power On This Virtual Machine

Install Ubuntu/Windows.

Step 9: Observe Performance

Ubuntu

Open Terminal

top

or

free -h

Windows

Open

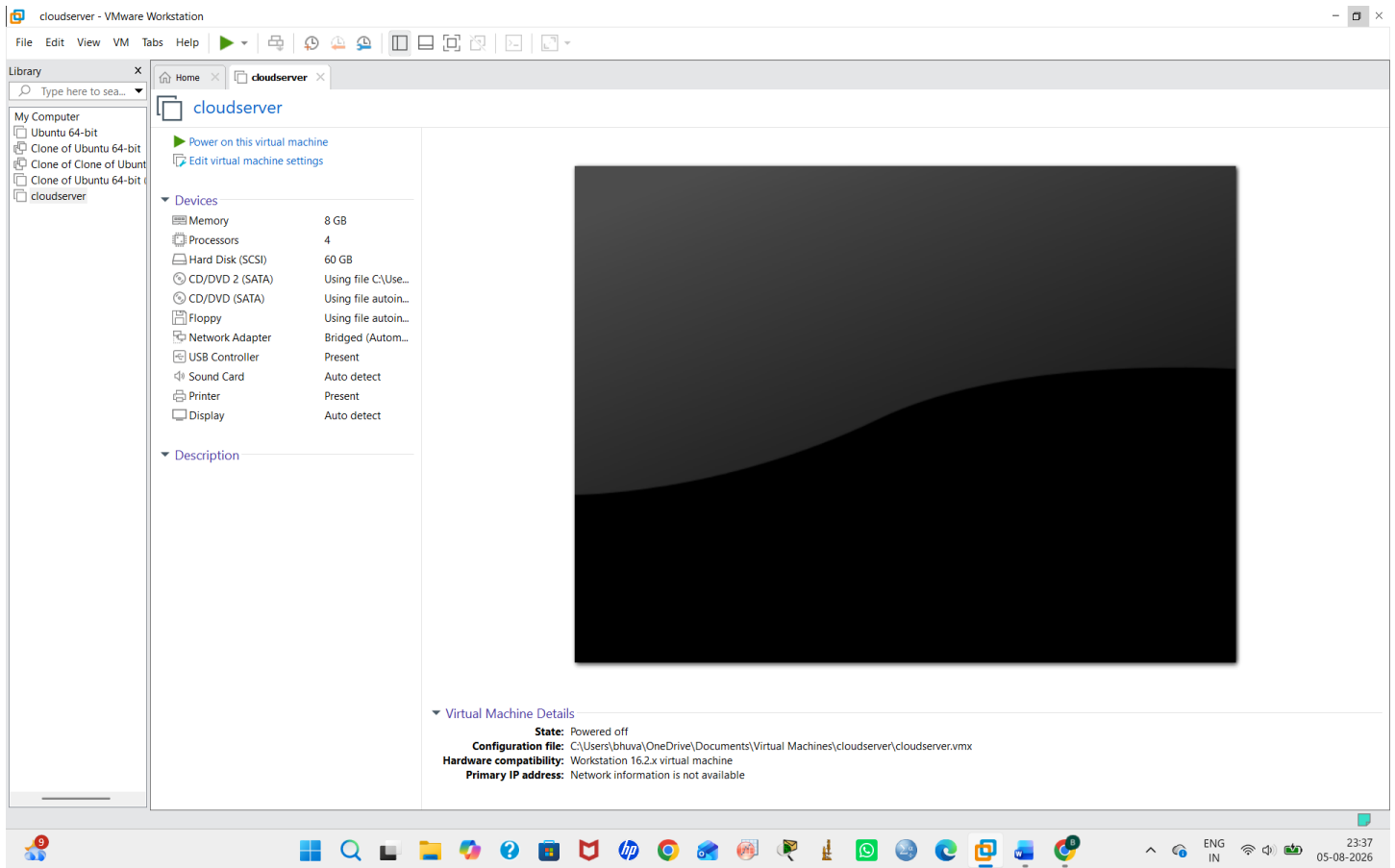
Task Manager

Observe

CPU

Memory

Disk



Question 2: Isolation of Finance, Healthcare and Student VMs

Step 1

Create three VMs

Finance VM

Healthcare VM

Student VM

You can clone the first VM.

VM

Manage

Clone

Choose

Full Clone

Repeat twice.

Step 2

Open
Edit

Virtual Network Editor
Click
Change Settings

Step 3

Create Networks
Add Network

VMnet2
Again
VMnet3

VMnet4
Example
VMnet2
192.168.10.0

VMnet3
192.168.20.0

VMnet4
192.168.30.0

Step 4

Finance VM
Settings

Network Adapter

Custom

VMnet2
Healthcare
VMnet3
Student
VMnet4

Step 5

Assign Static IP
Finance
192.168.10.10

Healthcare
192.168.20.10
Student
192.168.30.10

Step 6

Ping Test

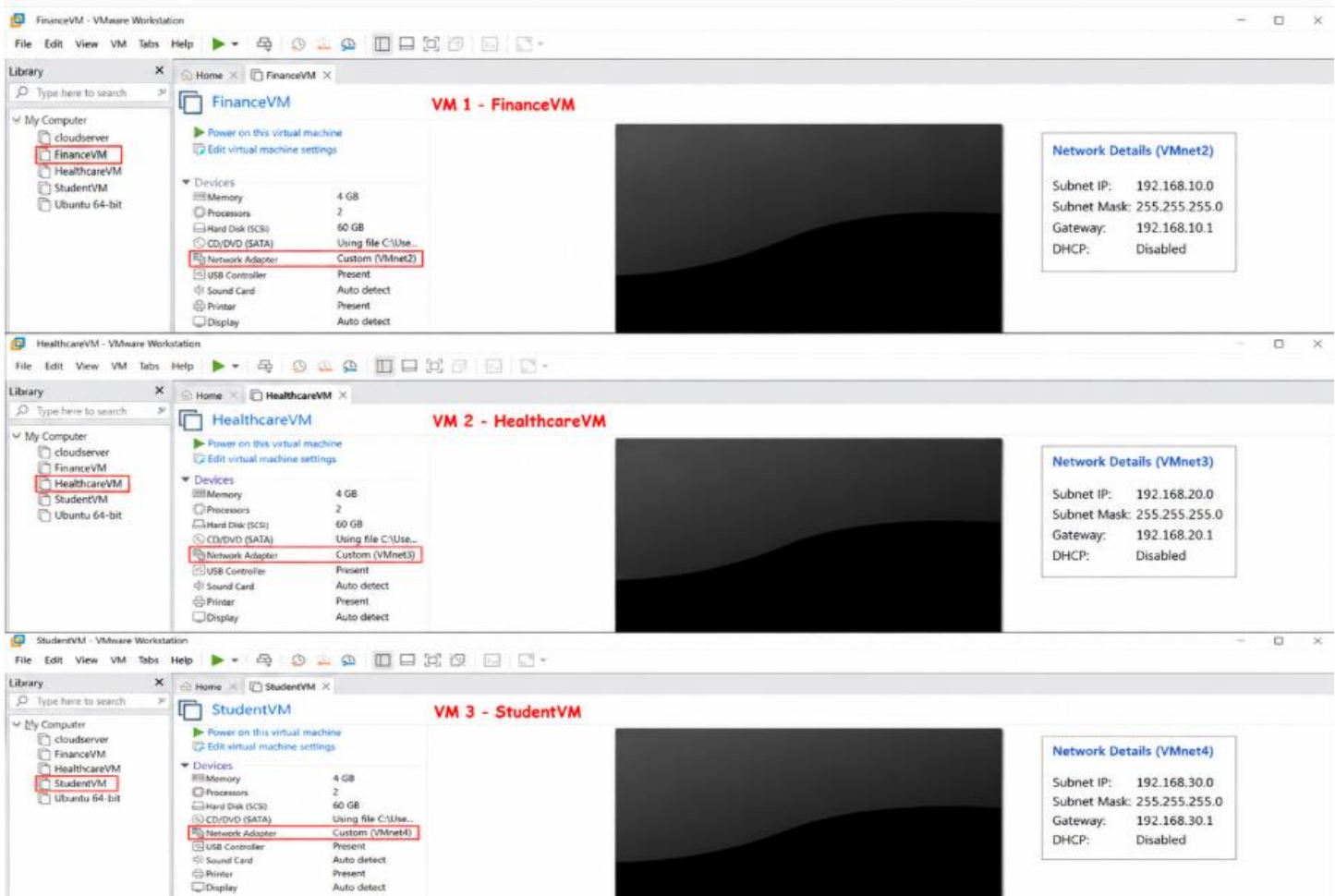
Finance VM

ping 192.168.20.10

Expected

Request timed out

This proves isolation



Question 3: Host Utilization

Given
CPU 92%

Memory 88%

Storage Latency 35 ms

VM Boot Time 170 sec

Step 1

Shutdown VM

Step 2

Increase RAM
VM

Settings

Memory

4096 MB

↓

8192 MB

Step 3

Increase CPU
Processors

2

↓

4

Step 4

Clean Disk
VM

Manage

Clean Up Disks

Wait until completed.

Step 5

Compact Disk
VM

Manage

Clean Up Disks

Compact

Step 6

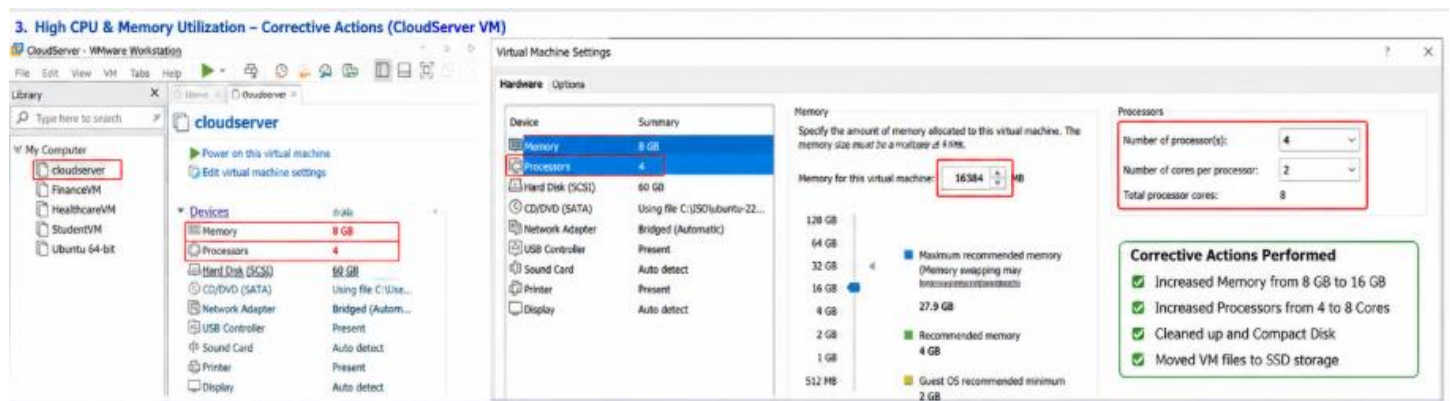
Move VM to SSD
If stored on HDD
Move
VM Folder



SSD

Step 7

Start VM
Measure boot time again.



Question 4: Software Defined Datacenter

Create three VMs.

Ubuntu Compute

Ubuntu Storage

pfSense

Step 1

Create Ubuntu Compute VM.

Step 2

Clone

Ubuntu Storage

Step 3

Create pfSense VM.

Step 4

Open

Edit

Virtual Network Editor

Create

VMnet5

Step 5

Attach all VMs

Settings

Network Adapter

Custom

VMnet5

Step 6

Take Snapshot

VM

Snapshot

Take Snapshot

Name

InitialState

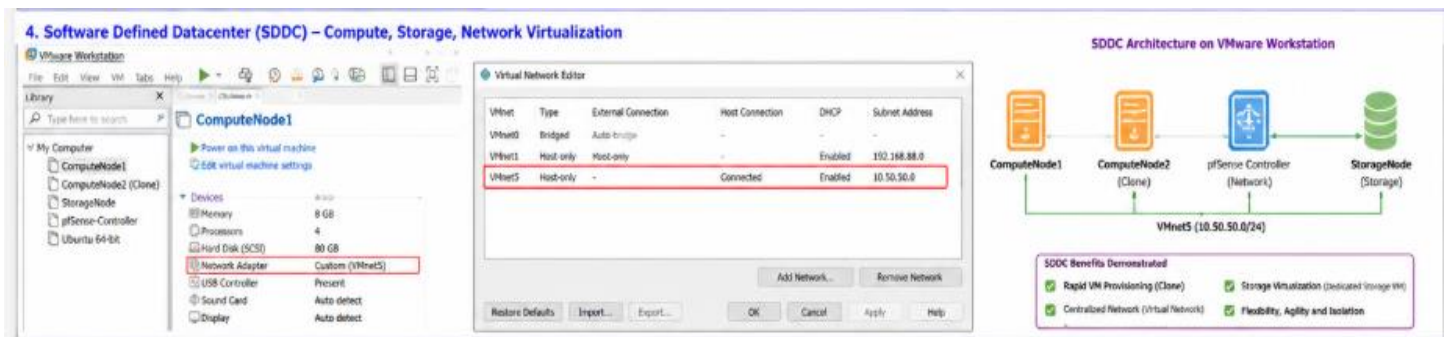
Step 7

Clone Compute VM
VM

Manage

Clone

This demonstrates rapid provisioning.



Question 5: Identity Federation

Use Ubuntu Server.

Step 1

Create Ubuntu Server VM.

Memory

4 GB

Disk

30 GB

Step 2

Install Docker

Open Terminal

`sudo apt update`

`sudo apt install docker.io -y`

Step 3

Start Docker

`sudo systemctl enable docker`

`sudo systemctl start docker`

Step 4

Download Keycloak

```
docker run -d \  
-p 8080:8080 \  
-e KEYCLOAK_ADMIN=admin \  
-e KEYCLOAK_ADMIN_PASSWORD=admin \  
quay.io/keycloak/keycloak:latest \  
start-dev
```

Step 5

Open Browser

<http://localhost:8080>

Login

Username

admin

Password

admin

Step 6

Create Realm

CloudFederation

Step 7

Create Users

finance

doctor

student

Step 8

Create Identity Provider

Choose

OpenID Connect

or

SAML

Step 9

Create Clients

Cloud-A

Cloud-B

Enable Single Sign-On

5. Identity Federation (Multi-Cloud) – Keycloak on Ubuntu Server

The screenshot displays a VMware Workstation environment with an Ubuntu VM named 'Ubuntu-IdP (Keycloak)'. The VM's hardware configuration is visible on the left, including 4 GB of memory and a bridged network adapter. The terminal window shows the installation of Keycloak via Docker and the configuration of a 'master' realm. The Keycloak Admin Console is open in a browser, showing the 'Cloud Federation' settings page. The 'Users' tab is active, displaying a table of users and a 'Federation Summary' section.

Terminal Output:

```
user@ubuntu-idp:~$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  STATUS    PORTS
d7f3be7ad2    quay.io/keycloak/keycloak:latest    "/opt/keycloak/bin..." Up 2 minutes    8.6.6.8:8080->8080/tcp    keycloak

user@ubuntu-idp:~$ curl http://localhost:8080/realms/master
{
  "realm": "master",
  "public_key": "-----BEGIN PUBLIC KEY-----",
  "token-service": "http://localhost:8080/realms/master/protocol/openid-connect/token",
  "endpoints": {
    "authorization": "http://localhost:8080/realms/master/protocol/openid-connect/auth",
    "token": "http://localhost:8080/realms/master/protocol/openid-connect/token",
    "userinfo": "http://localhost:8080/realms/master/protocol/openid-connect/userinfo"
  }
}
```

Keycloak Admin Console - Cloud Federation Settings:

Username	Enabled	Email	Actions
finance	true	finance@example.com	Edit
doctor	true	doctor@example.com	Edit
student	true	student@example.com	Edit

Federation Summary:

- Keycloak running on Ubuntu (Port 8080)
- Realm: CloudFederation created
- Users: finance, doctor, student added
- Ready for SAML / OpenID Connect federation with Cloud-A and Cloud-B